

FPGA - BASED HEALTH MONITORING WITH ML-BASED DECISION LOGIC

Tanuka Sree. R

SELECT

VIT Chennai

Chennai, India

tanukasree.r2023@vitstudent.ac.in

Karthik. K

SELECT

VIT Chennai

Chennai, India

karthik.k2023a@vitstudent.ac.in

Subramanian. N

SELECT

VIT Chennai

Chennai, India

subramanian.n2023@vitstudent.ac.in

Abstract – *The development of intelligent healthcare monitoring solutions is being driven by the fast-growing field of Embedded Systems and Real-time Computation, which has greatly impacted the process of designing and implementing an Embedded Healthcare Monitoring System. This Embedded System was developed on a Xilinx Artix-7 FPGA and is capable of evaluating a patient's health condition using multiple physiological parameters, such as Heart Rate, Oxygen Saturation (SpO₂), Body Temperature, Blood Pressure, and Respiratory Rate. Also, instead of traditional sensors for acquiring these data or software processing to interpret their information, the proposed system uses digital input (in the form of switch) to emulate real-time patient condition, thus, concentrating on creating an efficient hardware solution.*

To determine the order of importance for each of the aforementioned physiological factors, a machine learning (ML) model was created using a Random Forest Classifier to derive Feature Importance Values from a dataset collected in a realistic environment. The derived Feature Importance Values were converted and implemented as weights into hardware via the FPGA Logic. In addition to the static Use of Weights, the system is also able to incorporate a Temporal Persistence Mechanism that only accounts for abnormal conditions to the cumulative Risk Score after those conditions have persisted for a set number of clock cycles, thereby removing the effects of Transient Noise, and increasing the overall reliability of the system.

The cumulative Risk Score is computed continuously by the FPGA, based on the weighted inputs, and when the Cumulative Risk Score exceeds a predetermined Threshold, an Alert is triggered. The proposed solution will exhibit low latency, deterministic characteristics, and extreme scalability capabilities, thus providing a comprehensive platform for integrating ML-based insights to improve real-time support for clinical decision-making.

Key Words: *FPGA, Artix-7, Verilog, Health Monitoring, Machine Learning, Random Forest, Risk Scoring, Embedded Systems, Finite State Machine (FSM), SpO₂, Vital Signs, Real-Time Systems.*

INTRODUCTION:

The introduction of health monitoring has become an essential tool in healthcare, because of the growing number of chronic diseases. In today's society, many patients are diagnosed based on a lack of symptoms until later into their chronic disease state or even after they develop complications due to lack of symptom management and/ or treatment options. Traditional health monitoring methods require periodic monitoring and need physical intervention to be made before the patient can receive treatment. The advancements of embedded systems, Internet of Things (IoT), FPGA technology, have allowed for the creation of real-time, automated health monitoring solutions that have been rapidly gaining traction.

IoT-based health care monitoring has allowed continuous tracking of vital parameters and the transmission of data to cloud-based platforms, thus allowing for remote diagnosis of patients and the minimization of direct patient-doctor interaction. Wearing sensors and using wireless communication technology have enhanced the availability of healthcare services, particularly in remote areas; however, these systems often rely heavily on the processing of information through software and on connecting to the cloud, which can result in concerns regarding latency and reliability. The development of health monitoring systems based on FPGAs has gained popularity in recent years as a high-speed processing solution with the ability to perform many tasks simultaneously (parallelism) at predictable times (deterministic). A new health monitoring system incorporating the use of multiple sensors and the I2C communications protocol to communicate between sensor and patient equipment has been introduced to monitor heart rate, temperature, and oxygen saturation levels in real-time. The method utilized by this new health monitoring system includes processing of sensor data in real-time on the FPGA and sending alerts to healthcare professionals when the system detects abnormal cardiac activity (abnormality) providing an opportunity for immediate intervention and increased reliability. In addition to the ability to process multiple sensor inputs in real-time, this Modular design and efficient use of the I2C protocol for communication will create the opportunity for

seamless integration of the health monitoring system at both ends of the physician-patient spectrum. To overcome the limitations of current systems, we propose to develop a 100% fully hardware health-monitoring system that utilizes FPGAs, which will use digital signals to simulate physiological parameters, while employing machine-learning methodologies for deriving, evaluating, and determining the relative contribution of the parameters to the predicted outcome accordingly. An extensive decision-making system has been built with the inclusion of a multi-parameter scoring mechanism, temporal persistence logic, and a finite State Machine (FSM) control structure. It is anticipated that the combination of machine learning with an FPGA-based processor will provide a comprehensive solution for the next generation of Health Monitoring Systems (HMS).

The proposed hardware implementation will employ an Artix-7 FPGA (Xilinx) to achieve high performance with low power consumption. These FPGAs have a wide variety of configurable logic blocks (CLBS), configurable lookup tables (LUTs), flip-flops, and internal memory resources available to perform both combinatorial and sequential logic functions within a single device. In addition, Artix-7 FPGAs have general-purpose input/output (GPIO) capability to easily connect both switches (to generate input conditions) and LEDs (to indicate output conditions). This capability, along with the inherent parallelism of FPGAs, allows multiple functions to be executed simultaneously resulting in increased system efficiency (throughput) and lower latency.

The weights will be normalized and scaled to an integer value to ensure that they can be used in hardware implementations. Therefore, we will build a single system that is comprised of both FPGA hardware and ML software, which will allow for intelligent decision making in real-time.

The Human Vital Signs dataset will be used in this study. The dataset includes Physiological Parameters and Risk Category assignments for several types of physiological parameters (e.g. heart rate, respiratory rate, oxygen saturation, body temperature, systolic and diastolic blood pressure). Every record is labelled as either a high or low risk, allowing for supervised learning (classification) in developing this ML model.

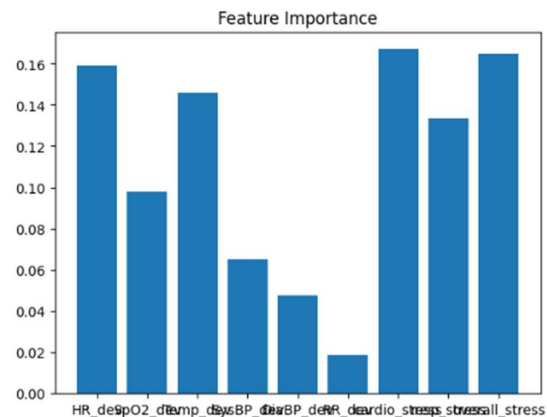


Fig. 1: The feature importance plot shows that combined stress indicators and heart rate contribute most to risk prediction, while individual parameters like RR and BP have relatively lower influence.

In order to facilitate the transfer of the data to the hardware, the continuous values will be converted to binary flags that represent whether a condition is abnormal or not, and those flags will be generated based upon thresholds determined through a data-driven methodology from the dataset, rather than the fixed clinical standard. This will allow the model to identify the relative abnormality of the parameters in the dataset, resulting in a more effective model for classification purposes. The resultant binary flags will then be used to train the ML model to extract the appropriate weights to use in the FPGA.

VIVADO DESIGN ARCHITECTURE:

The proposed FPGA based health monitoring system uses a modular hardware designing approach in the Vivado design suite. Each of the functional block in the schematic is created as an independent Verilog module and not connected through as a top level architecture. The overall architecture includes the following modules

- 1) Top module (Top.v) : This module integrates all the submodules and manages the full data flow while testing in the FPGA board.
- 2) Timer module (timer.v) : This module generates a periodic tick signal that corresponds to real-time seconds.
- 3) Input Handler module (input_handler.v) : This module ensures that all the asynchronous switch inputs are synchronised by a two stage flip flop synchronizer to avoid metastability.
- 4) Persistence Checker module (persistence_checker.v) : This module checks the input signals against the defined thresholds to remove transient noise.
- 5) Health Analyser module (health_analyzer.v): This module calculates the cumulative score based on the validated input and classifies the system status.
- 6) Output controller module (output_controller.v) : This module drives LEDs and a buzzer based on the system status that was classified in the health analyser module.
- 7) LCD controller (lcd_controller.v) : This module connects with a 2 X 16 LCD to show systems states (Safe, Warn, Help).
- 8) Testbench module (top_tb.v): This module verifies the functional correctness and robustness of the design system through simulation.

ALGORITHM:

The proposed system is based on a decision-making algorithm which validates the input signal and determines the status states based on the result of the validation and a severity classification.

The systems initially start with an idle state where all the outputs are reset. The system then monitors the input signal which corresponds to 8 different physiological parameters that include temperature, heart rate, SpO2, systolic blood pressure, Diastolic blood pressure, respiratory rate, ECG abnormalities and Emergency button.

The input signals are then synchronized in the input handler block followed by persistence validation. the persistence checker ensures that only those signals that are considered valid persists for a specified period, thereby avoiding false alarms during transient periods of abnormality.

Each of the physiological parameters are then provided with a weight that makes up the weighted score and altogether calculates the status of the condition based on the threshold value. According to this threshold value, the system state is classified as follows:

- 1) SAFE – No abnormality detected
- 2) WARNING – Abnormality detected
- 3) CRITICAL – Life threatening situation detected

An emergency input immediately forces the system into critical state regardless of other conditions. Based on the calculated states, the system generates appropriate outputs through LEDs, Buzzer alert and LCD displays.

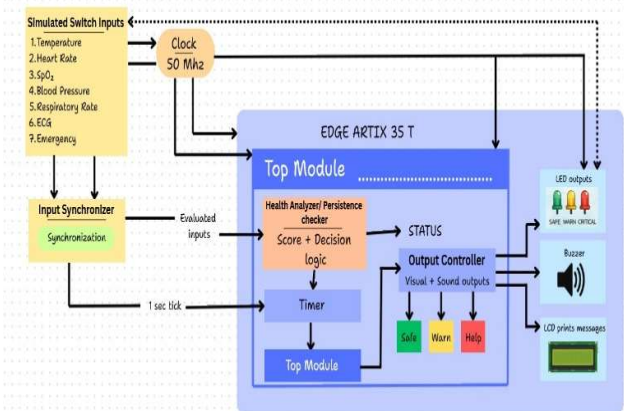


Fig. 2: Future Implementation Block Diagram

SCHEMATICS

The RTL schematic generated by vivado is shown below. This diagram explains the interconnections of all modules at hardware level. The top module (top.v) acts as a central unit where the input, processing and output modules are instantiated.

The use of this modular RTL structure provides clarity in the design, making it easier of efficient implementation.

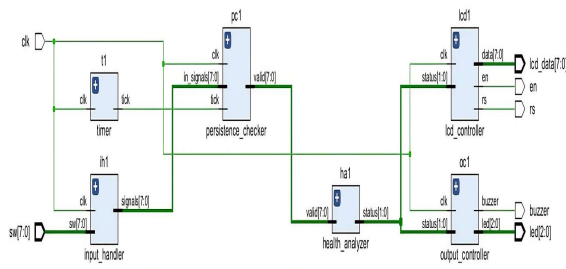


Fig. 3 : Schematic view of the designed system

SIMULATION AND TESTBENCH RESULTS:

The functional verification of the system is performed by simulating a behavioural testbench using Vivado software. This testbench contains various test cases that is the combination of inputs to observe the desired output.

KEY OBSERVATIONS FROM THE SIMULATION RESULTS:

- 1) Clock and tick signals are generated correctly
- 2) Signals are properly synchronized
- 3) The status of the system changes with respect to the validation done in the persistence checker module
- 4) Correct activation of LED and buzzer signals

SIMULATION WAVEFORMS:

- 1) INPUT = 0000 0000:
When the Input is 0000 0000, no abnormal signs or error is detected from any of the body parameters. This results in the weightage score of 0 indicating that the user is in a “SAFE STATE”. Accordingly, the system activates a GREEN LED and the Buzzer remains in the OFF condition.

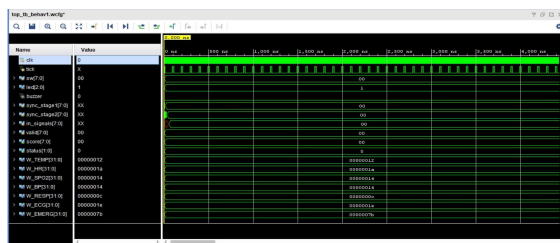


Fig. 4: Simulation waveform with input 00000000

- 2) INPUT = 0001 1000:
When the Input is 0001 1000, the abnormal signs in Systolic and diastolic pressure are detected from body. After the persistence delay, the weightage score is increased significantly indicating that the user is in “WARN STATE”. Accordingly, the system activates a

YELLOW LED, and the Buzzer turns into the OFF condition.

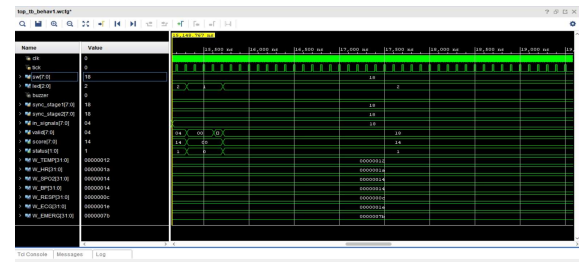


Fig. 5: Simulation waveform with input 00011000

- 3) INPUT = 1111 1111:
When the Input is 1111 1111, the abnormal sign is detected for all the parameters from the body. The weightage score is increased to the maximum indicating that the user is in “HELP STATE”. Accordingly, the system activates a RED LED, and the Buzzer turns into the ON condition.

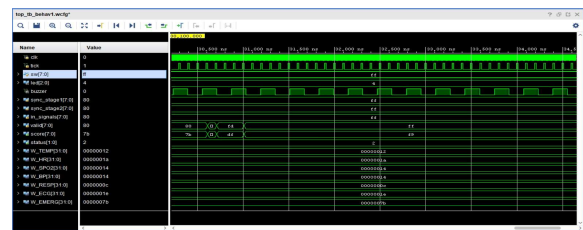


Fig. 6: Simulation waveform with input 11111111

HARDWARE IMPLEMENTATION

The designed system is implemented using an FPGA Development Board names “EDGE ARTIX-7 35T”. The hardware configuration of the board which is used in the proposed system includes:

1. FPGA Board
2. 2 X 16 LCD Display
3. On board switches
4. LEDs
5. 5V buzzer

Here, switches are utilised to mimic the activity of the physiological parameters and the resl time testing is done.

SYSTEM OPERATION:

The FPGA reads inputs from the on board switched and process them using the designed modules. According to the calculated state of the health condition, the system produces outputs as following:

- 1) The LED display system glows as per our IO port planning.

- 2) The Buzzer produces an alarm sound during critical state
- 3) The lcd displays text messages with respect to the health state.
- 4) The system operates in real-time and updates its outputs in accordance with the inputs

TEST CASE ANALYSIS

A series of test cases were conducted to validate the system.

- 1) TEST CASE 1: Normal condition
Input (Switches): 00000000
Observation:
 - (SAFE) LED is on
 - LCD displays “SAFE”
 - Buzzer is OFF

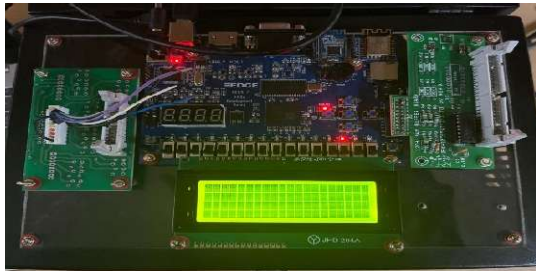


Fig. 7: User when in a Normal Condition

- 2) TEST CASE 2: Temperature Abnormality
Input (Switches): 00000001
Observation:
 - (Warning) LED is on
 - LCD displays “WARN”
 - Buzzer is OFF

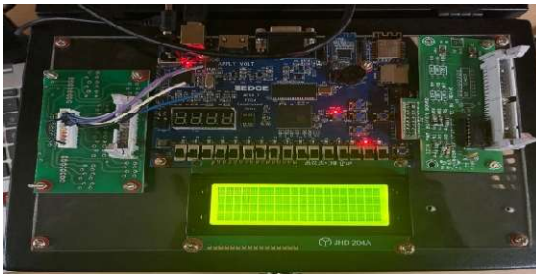


Fig. 8: User when in a Warning Condition

- 3) TEST CASE 3: SpO2 Critical Condition
Input: 00000100
Delay: ~4 seconds
Observation: after delay
 - (Critical) LED is ON.
 - LCD displays “HELP”
 - Buzzer is ON

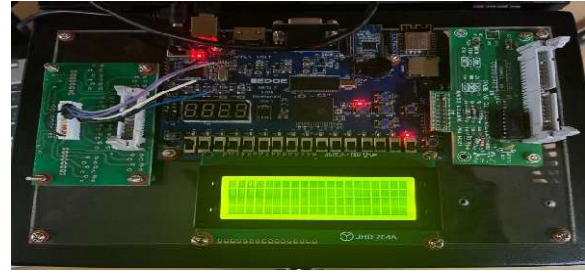


Fig. 9: User when in a Critical Condition

Hence the designed system will be able to:

1. Handle transient signals effectively by the addition of persistence
2. Be highly responsive in real-time to critical conditions.
3. Take decisions based on priority in emergency conditions
4. Integrate hardware and software effectively using FPGA

FUTURE IMPLEMENTATION PLAN:

For now, only the digital inputs are given to the system via the on-board switches. This cannot be related with the sophisticated medical field which requires real-time monitoring from the patient from the sensors. The system monitors many parameters and each of those can be given as an input via the sensor, i.e., sensors can directly be interfaced with the FPGA board, thresholds can be set for each of the parameter and can be decoded into digital high/low accordingly (see fig. below for block diagram implementation of sensor interfacing with the FPGA). Also, for each parameter input, a suitable sensor can be used to record the values real-time (see table).

Parameter	Measured Value	Sensor / Module	Output Type
Heart Rate (HR)	Pulse / BPM	MAX30102 / MAX30100	I2C (digital)
SpO ₂	Oxygen saturation	MAX30102 / MAX30100	I2C (digital)
Blood Pressure (BP)	Systolic & Diastolic	MPX5050 + cuff / Digital BP module	Analog / Serial
Respiratory Rate (RR)	Breathing pattern	Piezo belt sensor / Stretch sensor / IMU	Analog
Temperature	Body temperature	LM35 / DS18B20 / MAX30205	Analog / Digital
ECG	Electrical heart signal	AD8232 ECG module / BioAmp EXG Pill	Analog
Emergency	Manual trigger	Push Button Switch	Digital (GPIO)

Table. 1: Sensors that can be used for implementation